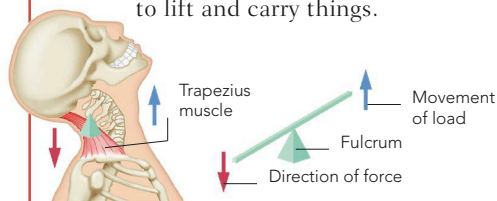


BODY PARTS AS LEVERS

Body movements employ the mechanical principles of applying a force to one part of a rigid lever, which tilts at a pivot point (fulcrum) to move a weight (load) that is elsewhere on the lever. The muscles apply force, bones serve as levers, and joints function as fulcrums. The various lever systems in the body allow a wide range of movement as well as making it possible to lift and carry things.

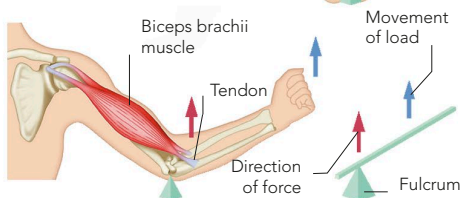
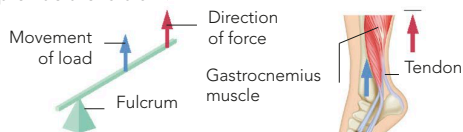


FIRST-CLASS LEVER

The fulcrum is positioned between the force and the load, like a see-saw. An example of this type of lever in the body is seen in the posterior neck muscles that tilt back the head on the cervical vertebrae.

SECOND-CLASS LEVER

The load lies between the force and the fulcrum. Standing on tip-toe, the calf muscles provide the force, the heel and foot form the lever, and the toes provide the fulcrum.

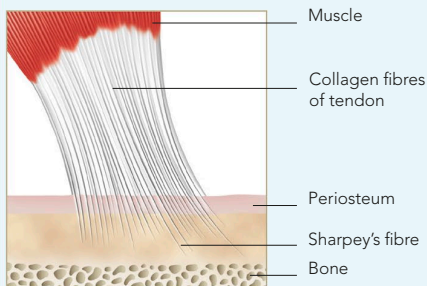


THIRD-CLASS LEVER

The most common type of lever in the body; the force is applied between the load and the fulcrum. An example is flexing the elbow joint (the fulcrum) by contracting the biceps brachii muscle.

TENDONS

Tendons are tough, fibrous cords of connective tissue that link skeletal muscles to bones. Within them, Sharpey's fibres pass through the bone covering (periosteum) to embed in the bone. Tendons in the hands and feet are enclosed in self-lubricating sheaths to protect them from rubbing against the bones. From the hand bones, tendons extend up to muscles near the elbow.



BONE-TENDON ATTACHMENT

Sharpey's fibres, which are also known as perforating fibres, are extensions of the tendon's proteinaceous collagen fibres.

